

RAPID run_id-centered redesign substrate

RAPID organizes redesign workflows around a persistent artifact layer. Multiple modules can read from and write to the same run-scoped artifacts, enabling resource-aware allocation of expensive structure-prediction budgets, safe reruns, backend replacement, and experimental-feedback evolution.



Design request

target • mask / position constraints • resource budgets • mode

Replaceable model modules

- MSA search (MMseqs2)
- Structure context (RFD3, BioEmu)
- Sequence design (ProteinMPNN)
- Solubility scorer / filter (SoluProt)
- Structure prediction (AF2 / ColabFold)
- Embedding / features (ESM-2)

Modules can be swapped without changing stored artifacts.

Run_id-centered artifact layer

Single source of truth for the run

Requests	input parameters, budgets, configurations	Surrogate labels	bootstrap labels, policy scores, CV results
Provenance / trace	stage events, logs, dependencies, versions	AF2 / ColabFold records	structures, pLDDT, optional relax scores, files
Stage outputs	MSA, backbones, designs, embeddings, scores, etc.	Selected candidates	acquisition lists, Top-K, selection rationale
Quality-control summaries	per-target and per-stage QC metrics	Experiment records	assay data, metadata, objectives
		Final summaries	run-level metrics, plots, tables

Operational capabilities

- Partial reruns from any stage
- Backend / model replacement
- Retrospective analysis
- Reproducibility & provenance
- Future preference-dataset accumulation

All analyses operate on the same run-scoped artifacts.



1) Structural-context exploration (candidate pool generation)

Analyzed separately from the surrogate-triage budget benchmark.



Structural-context arms:
original backbone, RFD3, BioEmu, RFD3+BioEmu

Purpose: broaden candidate-pool distribution and upper-tail candidates, not surrogate-model training

Used in structural-context ablation analysis (Figure 3).



2) Resource-aware surrogate triage (used in budget benchmark; original-backbone benchmark)



Low cost → High cost

Example budget (per target in main benchmark):
~10,000 designs → 30 bootstrap labels → 20 AF2 evaluations
(total 250 AF2 calls for 49,946 candidates across 5 targets)



3) Experimental-feedback evolution (DBTL loop)



Stores outcomes as experiment records and connects them to computational states for iterative redesign.



Automation / LIMS



Laboratory systems



Researchers & operators



Storage / Cloud (HPC, object store)



Read / write artifacts



Module can be replaced without changing artifacts



Supports reruns, re-analysis, and future reuse

Cost key

- Low / inexpensive
- Moderate
- High / GPU-heavy